Dong 2020 Prostate Cancer — analysis summary

RUN 4 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

Malignant cells split into two disconnected lineage lobes — adenocarcinoma (patients #1/#2/#5) and small-cell neuroendocrine NEPC (patient #6) — and never sample-integrate. Judged on the regenerated embedding across a θ sweep with per-neighborhood iLISI (max 4): iLISI stays ~1.2 and each lobe is a per-patient archipelago; n_pcs-invariant (20/30/40). Mixing appears only at θ=8 by destroying the embedding. Not mixed, not continuous → do not force a trajectory.

Gate 2 — module geography → NA

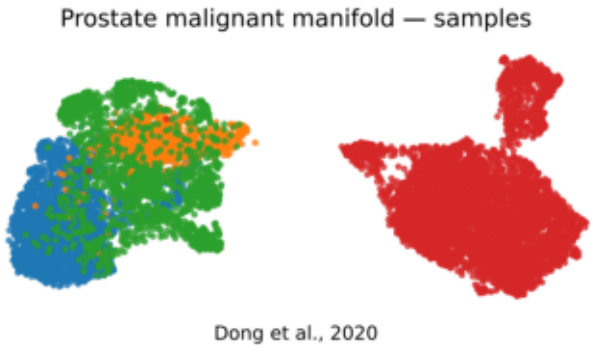
Not evaluated: the gate chain stops at the Gate 1 NO-GO (no manifold/trajectory is fit).

Gate 3 — hub cross-repression → NA

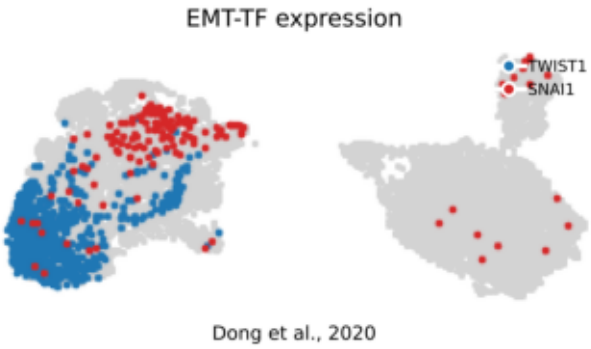
Not evaluated (branching-only; requires a fitted tree, which Gate 1 precludes).

STEP FIGURES

Manifold — samples



EMT-TF expression



Dataset.
Prostate cancer (Dong et al., 2020), 10x — 6 patients, 21,292 cells total; 9,698 malignant, of which 4,933 adenocarcinoma and 4,765 small-cell NEPC. Patients #3/#4 contribute no malignant cells.

Verdict.
The embryonic-program pattern does NOT reproduce. The malignant compartment is two distinct prostate-cancer lineages (adeno vs NEPC) that neither integrate nor form one continuous manifold; the adeno lobe alone is a per-patient archipelago. Gate 1 NO-GO — no trajectory fit. The EMT-TF program (TWIST1 dominant) is confined to the adeno lobe.

Parameter choices

Min-cell threshold	≥100 malignant/sample → keeps all 4 present samples (#3/#4 have 0)	**Overdispersed genes**	1,670 (scFates GAM)
Integration	Harmony(sample); θ sweep 0/2/5/8; n_pcs=20 (grid-invariant 20/30/40)	**Gate-1 metric**	per-neighborhood iLISI (k=90) + embedding read; mean iLISI 1.0/1.2/1.6/1.9 by θ (max 4)
Lobe biology	adeno: Gast–pSom r=+0.55, both oppose Crest; NEPC: distinct AR+/CHGA+ lineage	**Geometry**	none fit (Gate 1 NO-GO)
EMT-TFs	TWIST1 (dominant), SNAI1, FOXC1; VIM excluded		